

COMP7710

Programming Fundamentals

Today's lecture

Code Ethics and Professional Practice

Guest Lecturer
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Emma Davidson

- > Director of Information Management at Navy
- > Australia Day Achievement Award at Centrelink
- > Organiser of open source software conferences
- > ACT Government 2020 – 2024 Minister for Mental Health;
Minister for Disability; Minister for Justice Health and Corrections;
Minister for Population Health; Minister for Community Services,
Seniors & Veterans
- > Program Director for IDEATE



Agenda

- > Intellectual Property
- > Open source
- > Academic integrity
- > Professional conduct
- > Ethical leadership

Questions are welcome anytime!



Intellectual Property

- > Creations of the mind: inventions, literary or artistic works, designs, symbols, names
- > Protection through copyright, trademarks, patents, trade secrets
- > Ensures that creators of work are able to be financially compensated for their effort
- > Code is automatically protected by copyright as soon as you create it
- > Protection covers the code, but not the underlying ideas or methods. Someone else could independently create similar code using the same ideas and methods.



Who owns the code?

- > If you create code as part of your employment, your employer owns the code
- > If you create code as a self-employed contract worker, you own the code. But the terms of your contract with the company may require you to hand over IP rights to the code you create for that contract
- > If you create code as part of an open source project, even if you do so as part of your employment, the IP resides with either you or your employer (if you create the code on work time). But you grant rights under the open source license terms for the code to be used, modified and redistributed.



Open source licenses

- > MIT license: <https://opensource.org/license/MIT>
- > GNU v3: <https://opensource.org/license/gpl-3.0>
- > Apache 2.0: <https://opensource.org/license/apache-2.0>

- > Which license suits my project?
- > Balancing simplicity, protection, and method of use



Academic Integrity

- > Each course at ANU has its own rules for use of AI in creating code
- > Some courses do not allow the use of AI to generate code
- > If you use AI to generate code, comment the code and document its use
 - > What tool you used (eg Claude, Gemini, ChatGPT)
 - > How it was used
 - > The specific prompts you used
- > You are **not** the author for the code you generate using AI tools
- > You **are** responsible for the code you generate using AI tools and its effects
 - > Plagiarism
 - > Hallucinations
 - > Privacy
- > Breaking the rules is academic misconduct



Professional conduct

> ACS Code of Professional Conduct:

<https://www.acs.org.au/memberships/professional-ethics-conduct-and-complaints.html>

> Engineers Australia Code of Ethics:

<https://www.engineersaustralia.org.au/publications/code-ethics>

> IEEE Code of Ethics: <https://www.ieee.org/about/ethics>

> ACM Code of Ethics: <https://www.acm.org/code-of-ethics>



Ethical considerations

- > Code quality
- > Code ownership
- > Financial implications
- > Impacts on people and planet



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Ethical leadership

- > Be clear and consistent about your purpose
- > Be kind
- > Listen
- > Learn from experience, lead by example



Dhjan yimaba Thank you



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